

Group Theory

Basic Principles

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1 Groups

1.1 Definition and first properties

A **group** is a non-empty set G together with a binary operation

$$G \times G \longrightarrow G, \quad (g, h) \longmapsto gh,$$

satisfying the following axioms:

1. **Closure:** for all $g, h \in G$, the product gh is again an element of G .

2. **Associativity:** for all $g, h, k \in G$,

$$(gh)k = g(hk).$$

3. **Identity element:** there exists an element $e \in G$ such that, for all $g \in G$,

$$eg = ge = g.$$

4. **Inverse element:** for each $g \in G$, there exists an element $g^{-1} \in G$ such that

$$gg^{-1} = g^{-1}g = e.$$

The operation is often called **group multiplication**, even when it is not ordinary multiplication. If

$$gh = hg \quad \text{for all } g, h \in G,$$

then G is called an **abelian** or **commutative** group. Otherwise, G is called **non-abelian**.

The number of elements in G is called the **order** of G and is denoted by $|G|$. A group is **finite** if $|G|$ is finite; otherwise it is **infinite**.

Basic consequences

Let G be a group.

1. The identity element is unique.

2. The inverse of each element is unique.

3. For every $g \in G$, $(g^{-1})^{-1} = g$.

4. For every $g, h \in G$,

$$(gh)^{-1} = h^{-1}g^{-1}.$$

5. The cancellation laws hold: if $ag = ah$, then $g = h$; if $ga = ha$, then $g = h$.

1.2 Standard examples

The sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are infinite abelian groups under ordinary addition. The identity element is 0, and the inverse of x is $-x$.

The sets $\mathbb{Q}^\times$, $\mathbb{R}^\times$ and $\mathbb{C}^\times$ of non-zero rational, real and complex numbers are infinite abelian groups under ordinary multiplication. The identity element is 1, and the inverse of x is $1/x$.

The set $\{1, -1\}$ is an abelian group of order 2 under multiplication, and the set $\{1, -1, i, -i\}$ is an abelian group of order 4 under complex multiplication.

The set $M_n(\mathbb{R})$ of all $n \times n$ real matrices is an abelian group under matrix addition. The identity element is the zero matrix.

The set $GL(n, \mathbb{R})$ of all invertible $n \times n$ real matrices is a group under matrix multiplication. For $n \geq 2$, it is non-abelian.

1.3 Generators and presentations

A subset $S \subseteq G$ is said to **generate** a group G if every element of G can be written as a finite product of powers of elements of S . In this case, we write

$$G = \langle S \rangle.$$

If S contains only one element, then G is called a **cyclic group**.

A finite group can often be described by generators and relations. For example, the cyclic group of order n may be written as

$$C_n = \langle a : a^n = e \rangle = \{e, a, a^2, \dots, a^{n-1}\}.$$

The Klein four-group may be presented as

$$V = \langle a, b : a^2 = b^2 = e, ab = ba \rangle = \{e, a, b, ab\}.$$

These examples follow the generator-and-relation style used in the reference text.

1.4 Permutation groups

Let

$$S = \{a_1, a_2, \dots, a_n\}$$

be a set with n elements. A **permutation** of S is a bijective map

$$\sigma : S \longrightarrow S.$$

The set of all permutations of S forms a group under composition of maps. This group is called the **symmetric group** and is denoted by S_n . Its order is

$$|S_n| = n!.$$

A convenient notation for a permutation is the two-line notation

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix}.$$

For example, in S_3 ,

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

means $1 \mapsto 2$, $2 \mapsto 3$ and $3 \mapsto 1$. This is also written as the cycle (123) .

The group S_3 is the smallest non-abelian group. For instance, if

$$\alpha = (12), \quad \beta = (23),$$

then

$$\alpha\beta = (123), \quad \beta\alpha = (132),$$

so $\alpha\beta \neq \beta\alpha$.

1.5 Matrix groups

Matrix groups play an important role in physics because they describe rotations, unitary transformations and continuous symmetries.

The **general linear group** is

$$GL(n, \mathbb{F}) = \{A \in M_n(\mathbb{F}) : \det A \neq 0\},$$

where $\mathbb{F}$ is usually $\mathbb{R}$ or $\mathbb{C}$. The operation is matrix multiplication.

The **special linear group** is

$$SL(n, \mathbb{F}) = \{A \in GL(n, \mathbb{F}) : \det A = 1\}.$$

The **orthogonal group** is

$$O(n) = \{A \in GL(n, \mathbb{R}) : A^T A = I_n\}.$$

Its elements preserve the Euclidean inner product. The **special orthogonal group** is

$$SO(n) = \{A \in O(n) : \det A = 1\}.$$

The group $SO(2)$ describes rotations in the plane, while $SO(3)$ describes proper rotations in three-dimensional space.

The **unitary group** is

$$U(n) = \{U \in GL(n, \mathbb{C}) : U^\dagger U = I_n\},$$

where $U^\dagger$ is the conjugate transpose of U . The **special unitary group** is

$$SU(n) = \{U \in U(n) : \det U = 1\}.$$

The groups $SU(2)$ and $SU(3)$ are especially important in quantum mechanics and particle physics.

1.6 Cyclic groups and order of an element

Let $g \in G$. The powers of g are defined by

$$g^0 = e, \quad g^n = \underbrace{gg \cdots g}_{n \text{ factors}}, \quad g^{-n} = (g^{-1})^n.$$

The set

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$$

is a subgroup of G , called the **cyclic subgroup generated by g** . If $G = \langle g \rangle$ for some $g \in G$, then G is called a **cyclic group**, and g is called a **generator** of G .

The **order of an element** $g \in G$, denoted $o(g)$, is the smallest positive integer m such that

$$g^m = e.$$

If no such positive integer exists, then g has infinite order.

Example

In the multiplicative group $\{1, -1, i, -i\}$, the element i has order 4 because

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1.$$

Therefore

$$\{1, -1, i, -i\} = \langle i \rangle.$$

1.7 Further finite groups

The **quaternion group** is an eight-element non-abelian group that may be written as

$$Q = \{e, s, i, j, k, si, sj, sk\},$$

with defining relations

$$s^2 = e, \quad i^2 = j^2 = k^2 = ijk = s.$$

It is useful in mathematical physics because quaternions are closely related to rotations and spin.

The **dihedral group** D_n is the symmetry group of a regular n -gon. It has $2n$ elements and can be generated by a rotation r and a reflection s satisfying

$$r^n = s^2 = e, \quad sr = r^{-1}s.$$

Thus

$$D_n = \{e, r, r^2, \dots, r^{n-1}, s, sr, sr^2, \dots, sr^{n-1}\}.$$

The dihedral groups are central examples for understanding geometric symmetries.

1.8 Multiplication tables

A finite group can be described by its **multiplication table** or **Cayley table**. For example, the group $\mathbb{Z}_2 = \{0, 1\}$ under addition modulo 2 has table

+	0	1
0	0	1
1	1	0

and the group $\{1, -1\}$ under multiplication has table

·	1	-1
1	1	-1
-1	-1	1

In a group table, each row and each column contains every group element exactly once. This follows from the cancellation laws.

1.9 Equivalence relations and partitions

An **equivalence relation** on a set X is a relation $\sim$ satisfying reflexivity, symmetry and transitivity. This means $x \sim x$ for all $x \in X$; if $x \sim y$, then $y \sim x$; and if $x \sim y$ and $y \sim z$, then $x \sim z$.

Each equivalence relation divides the set into disjoint equivalence classes. Conversely, any partition of a set defines an equivalence relation. This idea is useful in group theory because cosets, conjugacy classes and quotient groups are all built from equivalence relations.

1.10 Conjugation and conjugacy classes

Let G be a group and let $a, g \in G$. The element

$$gag^{-1}$$

is called the **conjugate** of a by g . Two elements $a, b \in G$ are **conjugate** if there exists $g \in G$ such that

$$b = gag^{-1}.$$

Conjugacy is an equivalence relation on G . The equivalence class of a is called the **conjugacy class** of a :

$$\mathcal{C}(a) = \{gag^{-1} : g \in G\}.$$

In an abelian group, every conjugacy class contains exactly one element, since

$$gag^{-1} = agg^{-1} = a.$$

For non-abelian groups, conjugacy classes can contain several elements and give important information about the structure of the group.

2 Subgroup

2.1 Definition of subgroup

Let G be a group with identity element e . A non-empty subset $H \subseteq G$ is called a **subgroup** of G if H is itself a group under the operation inherited from G . We write

$$H \leq G.$$

This section does not introduce new groups from the beginning; instead, it studies smaller groups that live inside a given group.

The subgroups $\{e\}$ and G are called **trivial subgroups**. A subgroup H is called **proper** if

$$H \neq G.$$

A proper subgroup may still be non-trivial if $H \neq \{e\}$.

2.2 Subgroup test

To prove that a subset $H \subseteq G$ is a subgroup, we do not need to check all group axioms again. Since associativity is already inherited from G , it is enough to check closure and inverses.

A useful one-step test is

$$H \neq \emptyset \quad \text{and} \quad a, b \in H \Rightarrow ab^{-1} \in H.$$

If this condition holds, then H contains the identity element, contains inverses, and is closed under the group operation. Hence $H \leq G$.

For finite subsets, the test becomes even simpler. If H is a non-empty finite subset of G and

$$a, b \in H \Rightarrow ab \in H,$$

then H is a subgroup. Finiteness forces the inverse of each element to occur among its powers.

2.3 Examples and non-examples

The set

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

is a subgroup of $(\mathbb{Z}, +)$ for every integer $n \geq 0$. These are exactly the subgroups of the additive group $\mathbb{Z}$.

Inside $GL(n, \mathbb{F})$, the special linear group

$$SL(n, \mathbb{F}) = \{A \in GL(n, \mathbb{F}) : \det A = 1\}$$

is a subgroup. If $A, B \in SL(n, \mathbb{F})$, then

$$\det(AB^{-1}) = \det(A) \det(B^{-1}) = 1.$$

Therefore $AB^{-1} \in SL(n, \mathbb{F})$.

The subset of positive real numbers $\mathbb{R}_{>0}$ is a subgroup of $\mathbb{R}^\times$ under multiplication. However, the interval $(0, 1)$ is not a subgroup, because the inverse of $1/2$ is 2 , which is not in $(0, 1)$.

The set of rotations in the plane is a subgroup of the group of all symmetries of the plane. The set of reflections alone is not a subgroup, because the product of two reflections is a rotation, not a reflection.

2.4 Subgroups generated by a set

If S is a subset of a group G , the **subgroup generated by** S is the smallest subgroup of G containing every element of S . It is denoted by

$$\langle S \rangle.$$

Equivalently, $\langle S \rangle$ consists of all finite products of elements of S and their inverses.

When $S = \{g\}$ has only one element, we write $\langle g \rangle$. This notation already appeared in Section 1 for cyclic groups, but here the emphasis is different: $\langle g \rangle$ is studied as a subgroup of a larger group G .

For example, in the symmetric group S_3 , the transposition (12) generates the subgroup

$$\langle (12) \rangle = \{e, (12)\}.$$

On the other hand, the two transpositions (12) and (23) generate all of S_3 :

$$\langle (12), (23) \rangle = S_3.$$

2.5 Cosets

Let $H \leq G$ and let $g \in G$. The **left coset** of H determined by g is

$$gH = \{gh : h \in H\},$$

and the **right coset** of H determined by g is

$$Hg = \{hg : h \in H\}.$$

Cosets are usually not subgroups, because they may not contain the identity element. Instead, cosets are translated copies of a subgroup.

Two left cosets of H are either equal or disjoint. Hence the left cosets of H partition G . The same statement holds for right cosets.

The number of distinct left cosets of H in G is called the **index** of H in G and is denoted by

$$[G : H].$$

For example, in $(\mathbb{Z}, +)$, the subgroup $2\mathbb{Z}$ has two cosets:

$$2\mathbb{Z} \quad \text{and} \quad 1 + 2\mathbb{Z}.$$

They are the even and odd integers, so $[\mathbb{Z} : 2\mathbb{Z}] = 2$.

2.6 Lagrange's theorem

If G is a finite group and $H \leq G$, then the order of H divides the order of G . More precisely,

$$|G| = [G : H]|H|.$$

This result is called **Lagrange's theorem**. It follows because the cosets of H partition G , and every coset has the same number of elements as H .

One consequence is that the order of every element $g \in G$ divides $|G|$, because the order of g is the number of elements in the subgroup $\langle g \rangle$.

Another consequence is that if $|G| = p$ is prime, then G has only two subgroups, namely $\{e\}$ and G . Hence every group of prime order is cyclic.

The converse of Lagrange's theorem is not always true. If a number d divides $|G|$, it is not guaranteed that G has a subgroup of order d .

2.7 Normal subgroups

A subgroup $N \leq G$ is called a **normal subgroup** if it is stable under conjugation:

$$gNg^{-1} = N \quad \text{for all } g \in G.$$

We write

$$N \trianglelefteq G.$$

Equivalently, N is normal if every left coset of N is also a right coset:

$$gN = Ng \quad \text{for all } g \in G.$$

Every subgroup of an abelian group is normal, because the equation $gn = ng$ holds for all $g \in G$ and $n \in N$. In a non-abelian group, subgroups need not be normal.

Normal subgroups are important because they allow us to construct quotient groups. If $N \trianglelefteq G$, then the set of cosets

$$G/N = \{gN : g \in G\}$$

forms a group under the operation

$$(gN)(hN) = (gh)N.$$

2.8 Center and centralizer

The **center** of a group G is the set of elements that commute with every element of G :

$$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}.$$

The center is always a normal subgroup of G . If G is abelian, then $Z(G) = G$.

For a fixed element $a \in G$, the **centralizer** of a is

$$C_G(a) = \{g \in G : ga = ag\}.$$

The centralizer is a subgroup of G . It measures how much of the group commutes with the element a .

3 Homomorphism

3.1 Definition

Let $(G, \cdot)$ and $(H, *)$ be two groups. A function

$$\varphi : G \longrightarrow H$$

is called a **group homomorphism** if it preserves the group operation, that is,

$$\varphi(g_1g_2) = \varphi(g_1) * \varphi(g_2) \quad \text{for all } g_1, g_2 \in G.$$

When the operation in both groups is clear, we simply write

$$\varphi(g_1g_2) = \varphi(g_1)\varphi(g_2).$$

A homomorphism is a map between groups that respects their algebraic structure.

3.2 Basic properties

If $\varphi : G \rightarrow H$ is a group homomorphism and e_G, e_H are the identity elements of G and H , then

$$\varphi(e_G) = e_H.$$

Indeed, since $e_G e_G = e_G$, we have $\varphi(e_G)\varphi(e_G) = \varphi(e_G)$, and cancellation in H gives $\varphi(e_G) = e_H$.

For every $g \in G$, a homomorphism also satisfies

$$\varphi(g^{-1}) = \varphi(g)^{-1}.$$

This follows because $gg^{-1} = e_G$, so $\varphi(g)\varphi(g^{-1}) = e_H$.

More generally, for every integer $n \in \mathbb{Z}$,

$$\varphi(g^n) = \varphi(g)^n.$$

Thus homomorphisms preserve powers and, therefore, preserve cyclic subgroup structure.

3.3 Kernel and image

The **kernel** of a homomorphism $\varphi : G \rightarrow H$ is the set of elements of G that are mapped to the identity element of H :

$$\ker \varphi = \{g \in G : \varphi(g) = e_H\}.$$

The kernel measures how many elements of G become invisible under the map.

The **image** of φ is the set of elements in H that are reached by elements of G :

$$\text{Im } \varphi = \{\varphi(g) : g \in G\}.$$

The kernel is a normal subgroup of G , and the image is a subgroup of H :

$$\ker \varphi \trianglelefteq G, \quad \text{Im } \varphi \leq H.$$

3.4 Injective and surjective homomorphisms

A homomorphism $\varphi : G \rightarrow H$ is **injective** if different elements of G have different images in H . For group homomorphisms, injectivity can be tested by the kernel:

$$\varphi \text{ is injective} \iff \ker \varphi = \{e_G\}.$$

This criterion is often easier than checking injectivity directly.

A homomorphism is **surjective** if every element of H is the image of some element of G . Equivalently,

$$\varphi \text{ is surjective} \iff \text{Im } \varphi = H.$$

A bijective homomorphism is called an **isomorphism**.

3.5 Isomorphisms

Two groups G and H are called **isomorphic**, written

$$G \cong H,$$

if there exists a bijective homomorphism $\varphi : G \rightarrow H$. Isomorphic groups have the same group structure, even if their elements are written differently.

For example, the additive group $\mathbb{Z}_2 = \{0, 1\}$ is isomorphic to the multiplicative group $\{1, -1\}$. One isomorphism is

$$0 \mapsto 1, \quad 1 \mapsto -1.$$

Addition modulo 2 corresponds exactly to multiplication of signs.

3.6 First isomorphism theorem

Let $\varphi : G \rightarrow H$ be a group homomorphism. Then the quotient group $G/\ker \varphi$ is isomorphic to the image of φ :

$$G/\ker \varphi \cong \text{Im } \varphi.$$

This theorem says that every homomorphism can be understood by first collapsing its kernel and then identifying the result with its image.

3.7 Cayley's theorem

Every finite group is isomorphic to a subgroup of a symmetric group. More precisely, if G is a finite group of order n , then G is isomorphic to a subgroup of S_n . This result is called **Cayley's theorem**.

The idea is that each element $g \in G$ defines a permutation of the elements of G by left multiplication:

$$L_g(x) = gx.$$

The map $g \mapsto L_g$ preserves products, because

$$L_g L_h = L_{gh}.$$

Therefore the group can be represented faithfully by permutations of its own elements.

4 Direct product

4.1 Cartesian product of groups

Let G and H be groups. The **direct product** of G and H is the Cartesian product

$$G \times H = \{(g, h) : g \in G, h \in H\}$$

with componentwise multiplication

$$(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 h_2).$$

This operation makes $G \times H$ into a group.

The identity element of $G \times H$ is

$$(e_G, e_H),$$

and the inverse of (g, h) is

$$(g, h)^{-1} = (g^{-1}, h^{-1}).$$

Thus all group operations are performed separately in each component.

4.2 Order of a direct product

If G and H are finite groups, then

$$|G \times H| = |G||H|.$$

This follows from the counting rule for Cartesian products. For example, if $|G| = m$ and $|H| = n$, then $G \times H$ contains mn ordered pairs.

If both G and H are abelian, then $G \times H$ is also abelian. Indeed,

$$(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2) = (g_2g_1, h_2h_1) = (g_2, h_2)(g_1, h_1).$$

If at least one factor is non-abelian, then the direct product is non-abelian.

4.3 Examples

The group $\mathbb{Z}_2 \times \mathbb{Z}_2$ has four elements:

$$(0, 0), (1, 0), (0, 1), (1, 1).$$

The operation is addition component by component modulo 2. This group is called the **Klein four-group**.

The group $SO(3) \times U(1)$ appears naturally when two independent symmetry operations are considered together. One component may describe rotations in space, while the other may describe a phase symmetry.

The group $SU(3) \times SU(2) \times U(1)$ is important in particle physics because it is the gauge symmetry group of the Standard Model. In this product, each factor describes a different type of internal symmetry.

4.4 Projection maps

There are natural homomorphisms from a direct product to its factors. These are called **projection maps**:

$$\pi_G : G \times H \longrightarrow G, \quad \pi_G(g, h) = g,$$

and

$$\pi_H : G \times H \longrightarrow H, \quad \pi_H(g, h) = h.$$

Both maps are surjective homomorphisms. They simply forget one component of the ordered pair.

4.5 Internal direct products

Sometimes a group K contains two subgroups G and H whose elements combine to describe every element of K . In this case, K may be written as an **internal direct product**.

A group K is the internal direct product of subgroups G and H if every element of K can be written uniquely as gh with $g \in G$ and $h \in H$, the subgroups commute elementwise, and their intersection is trivial:

$$G \cap H = \{e\}.$$

Under these conditions, K is isomorphic to the external direct product $G \times H$.

5 Vector spaces

5.1 Definition

Let $\mathbb{F}$ be a field, usually $\mathbb{R}$ or $\mathbb{C}$. A **vector space** over $\mathbb{F}$ is a non-empty set V whose elements are called vectors, together with vector addition and scalar multiplication. These operations are written as

$$u + v \in V, \quad \lambda v \in V,$$

where $u, v \in V$ and $\lambda \in \mathbb{F}$.

The operation of addition makes V an abelian group. Scalar multiplication must be compatible with addition and multiplication in the field. For example,

$$\lambda(u + v) = \lambda u + \lambda v, \quad (\lambda + \mu)v = \lambda v + \mu v, \quad (\lambda\mu)v = \lambda(\mu v), \quad 1v = v.$$

These axioms express the familiar rules for vectors.

5.2 Examples

The set $\mathbb{R}^n$ is a vector space over $\mathbb{R}$. Its elements are ordered n -tuples

$$(x_1, x_2, \dots, x_n),$$

and addition and scalar multiplication are performed component by component.

The set $\mathbb{C}^n$ is a vector space over $\mathbb{C}$. It may also be regarded as a vector space over $\mathbb{R}$, but then its dimension doubles.

The set $M_{m \times n}(\mathbb{F})$ of all $m \times n$ matrices over $\mathbb{F}$ is a vector space under matrix addition and scalar multiplication.

The set of polynomials with coefficients in $\mathbb{F}$ is a vector space. The subset of polynomials of degree at most n is also a vector space and is denoted by $\mathbb{F}_n[x]$.

5.3 Subspaces

A non-empty subset $W \subseteq V$ is called a **subspace** of V if W is itself a vector space under the same operations as V . To check this, it is enough to verify that for all $u, v \in W$ and all scalars $\lambda \in \mathbb{F}$,

$$u + v \in W, \quad \lambda u \in W.$$

Thus a subspace is closed under addition and scalar multiplication.

For example, a line through the origin in $\mathbb{R}^2$ is a subspace, but a line not passing through the origin is not a subspace. This is because every subspace must contain the zero vector.

5.4 Linear combinations and span

Let $v_1, v_2, \dots, v_k \in V$. A **linear combination** of these vectors is an expression of the form

$$a_1 v_1 + a_2 v_2 + \dots + a_k v_k,$$

where $a_1, a_2, \dots, a_k \in \mathbb{F}$.

The set of all linear combinations of $v_1, v_2, \dots, v_k$ is called their **span**:

$$\text{span}\{v_1, v_2, \dots, v_k\} = \{a_1 v_1 + a_2 v_2 + \dots + a_k v_k : a_i \in \mathbb{F}\}.$$

The span is always a subspace of V .

5.5 Linear independence and basis

Vectors $v_1, v_2, \dots, v_k$ are called **linearly independent** if

$$a_1 v_1 + a_2 v_2 + \dots + a_k v_k = 0 \quad \Rightarrow \quad a_1 = a_2 = \dots = a_k = 0.$$

If there is a non-trivial solution, then the vectors are called **linearly dependent**.

A **basis** of a vector space V is a linearly independent set that spans V . Once a basis is chosen, every vector in V can be written uniquely as a linear combination of basis vectors.

The number of vectors in a basis is called the **dimension** of V . For example,

$$\dim \mathbb{R}^n = n, \quad \dim M_{m \times n}(\mathbb{F}) = mn.$$

5.6 Linear maps

Let V and W be vector spaces over the same field $\mathbb{F}$. A function $T : V \rightarrow W$ is called a **linear map** if

$$T(u + v) = T(u) + T(v), \quad T(\lambda v) = \lambda T(v)$$

for all $u, v \in V$ and all $\lambda \in \mathbb{F}$.

Linear maps are the structure-preserving maps of vector spaces, just as homomorphisms are the structure-preserving maps of groups. The kernel and image of a linear map are defined by

$$\ker T = \{v \in V : T(v) = 0\}, \quad \operatorname{Im} T = \{T(v) : v \in V\}.$$

The kernel is a subspace of V , and the image is a subspace of W .

6 Tensor products

6.1 Motivation

Tensor products provide a way to combine vector spaces so that bilinear expressions become linear expressions. They are very useful in physics because composite systems, multilinear maps and representation spaces are naturally described using tensor products.

If V and W are vector spaces over the same field $\mathbb{F}$, then their tensor product is another vector space denoted by

$$V \otimes W.$$

The elementary vectors of this space are written as

$$v \otimes w,$$

where $v \in V$ and $w \in W$.

6.2 Bilinearity rules

The tensor product is defined so that it is linear in each component. This means

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w,$$

$$v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2,$$

and

$$(\lambda v) \otimes w = v \otimes (\lambda w) = \lambda(v \otimes w).$$

These identities are the most important computational rules for tensor products.

A general element of $V \otimes W$ is a finite sum

$$\sum_i v_i \otimes w_i.$$

Not every tensor can be written as a single elementary tensor $v \otimes w$; many tensors require sums of elementary tensors.

6.3 Basis and dimension

Suppose V has basis $\{e_1, e_2, \dots, e_m\}$ and W has basis $\{f_1, f_2, \dots, f_n\}$. Then $V \otimes W$ has basis

$$\{e_i \otimes f_j : 1 \leq i \leq m, 1 \leq j \leq n\}.$$

Therefore,

$$\dim(V \otimes W) = \dim(V) \dim(W).$$

This formula shows that tensor products multiply dimensions.

For example, if $V = \mathbb{R}^2$ and $W = \mathbb{R}^3$, then

$$\dim(V \otimes W) = 2 \cdot 3 = 6.$$

A basis is

$$e_1 \otimes f_1, e_1 \otimes f_2, e_1 \otimes f_3, e_2 \otimes f_1, e_2 \otimes f_2, e_2 \otimes f_3.$$

6.4 Tensor products of linear maps

Let $T : V \rightarrow V'$ and $S : W \rightarrow W'$ be linear maps. Their tensor product is the linear map

$$T \otimes S : V \otimes W \longrightarrow V' \otimes W'$$

defined on elementary tensors by

$$(T \otimes S)(v \otimes w) = T(v) \otimes S(w).$$

The definition then extends linearly to all finite sums of tensors.

This construction is important in representation theory. If a group acts on V and also acts on W , then it can act on $V \otimes W$ by acting on both components.

6.5 Tensor products in quantum mechanics

In quantum mechanics, the state space of a composite system is the tensor product of the state spaces of the individual systems. If one system has state space V and another has state space W , then the combined system has state space

$$V \otimes W.$$

A simple product state has the form

$$v \otimes w.$$

A general state may be a sum of product states, such as

$$v_1 \otimes w_1 + v_2 \otimes w_2.$$

Such states can describe entanglement, which cannot be represented by a single product tensor.

6.6 Relation with direct products

The direct product $V \times W$ and the tensor product $V \otimes W$ are different constructions. The direct product stores two vectors as an ordered pair

$$(v, w),$$

while the tensor product combines vectors into a new object

$$v \otimes w.$$

For finite-dimensional vector spaces,

$$\dim(V \times W) = \dim V + \dim W, \quad \dim(V \otimes W) = \dim V \dim W.$$

Thus direct products add dimensions, while tensor products multiply dimensions.